5.21×
BEST JOINT-CONTRACT FRONTIER
P99 1.45s · 8.2 J/audio-s

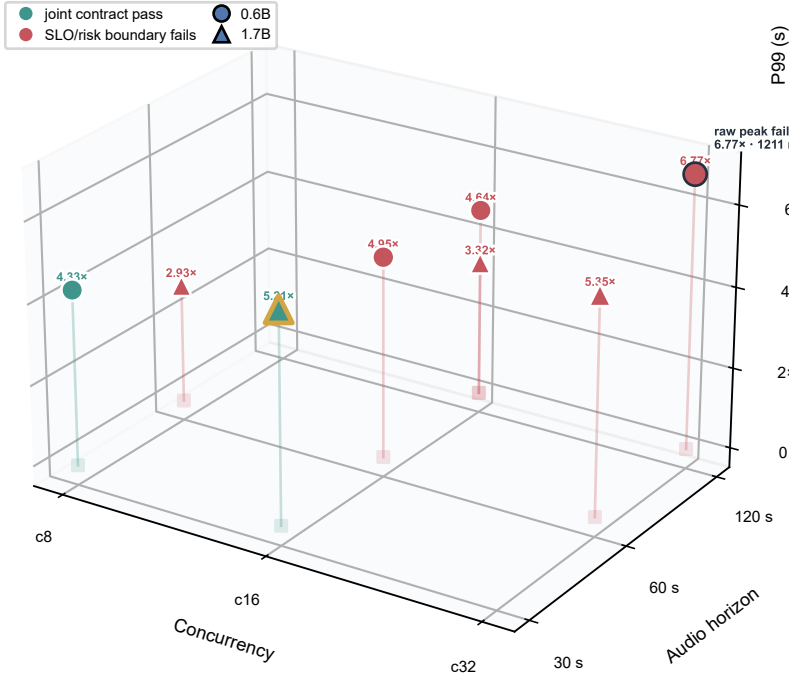
1.293s
RETURNED-CONTRACT P99
22.3 events/s · risk 0.448%

−95.29%
RECERTIFICATION EVENTS
1,358 → 64 exact-contract events

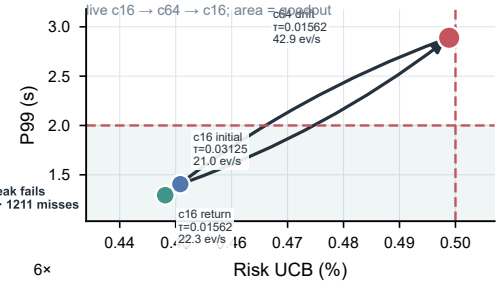
3 semantics
ARCHITECTURE COVERAGE
Qwen system · Whisper feedback · Voxtral append-only

a The 3D operating surface separates raw speed from safe speed

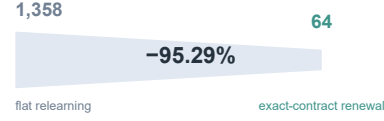
height = speedup · color = joint contract · marker = model · floor shadow = tested cell



b The controller follows the contract



c Renewal restores evidence cheaply



d Generality is layered, not overstated

system performance · causal transfer · append-only negative boundary

Qwen3-ASR · 0.6B/1.7B — SYSTEM
4.95–6.05× mean deployed speedup

Whisper · tiny/base — CAUSAL
8/21 ON → 0/21 OFF · protected: 0 violations

Whisper strong performance remains a 1.00× all-canonical boundary

Voxtral Realtime · 4B — BOUNDARY
append-only · 0/120 drift · bypass